

Saurabh Koju

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EDUCATION

- **Bachelors in Computer Engineering** November 2019 – March 2024
Pulchowk Campus, Institute of Engineering Lalitpur, Nepal
 - Rank 54 / 12,000 (**top 0.5%**) in national entrance exam.
 - Graduated in *First Division* with **78.97%**
- **Certifications**
 - **GRE:** 337 (Quant 170, Verbal 167)
 - **IELTS:** 8.5 (Listening 9.0, Reading 9.0, Writing 8.0, Speaking 7.5)

RESEARCH EXPERIENCE

- **Research Assistant** October 2024 - Present
Multimodal Learning Lab – Supervisor: [Dr. Binod Bhattarai](#) University of Aberdeen, UK & NAAMII
 - Working on research projects on **Surgical World Models**, **Medical Imaging** and **Generative Modeling**.
 - Introduced **surgical world model** based on inverse dynamics model that **learns latent actions from unlabeled surgical videos**.
 - Working on a novel method to obtain **latent actions that can be transferred across surgical contexts** with coarse latent action modeling.
 - Contributed to spatial proteomics image generation project with diffusion models.
- **Research Intern** July 2024 - September 2024
Multimodal Learning Lab – Supervisor: [Dr. Binod Bhattarai](#) University of Aberdeen, UK & NAAMII
 - **SAGES CVS Grand Challenge 2024:** *CVS Criteria Prediction in Cholecystectomy Videos, MICCAI 2024*
Developed a novel object-token transformer with relative layout embeddings that secured **Fourth position** in two sub-challenges: CVS Classification and Robustness.

PREPRINTS AND PUBLICATIONS

- **SurgTransfer: Multi-Step Latent Action Modeling in Surgical World Model.** *Under Review.*
Saurabh Koju, Love Panta, Deepak Thapa, Rudra P. K. Poudel, Prajwal Ghimire, Angus Watson, Yash Raj Shrestha, Prashanna Kumar Gyawali & Binod Bhattarai
- **The SAGES Critical View of Safety Challenge: A Global Benchmark for AI-Assisted Surgical Quality Assessment** *Under Review.* [Paper](#)
Deepak Alapatt, Jennifer Eckhoff, Zhiliang Lyu, ..., *Saurabh Koju*, ..., Nicolas Padoy
- **Surgical Vision World Model.** *Workshop on Data Engineering in Medical Imaging, MICCAI 2025.* [Paper](#)
Saurabh Koju, Saurav Bastola, Prashant Shrestha, Sanskar Amgain, Yash Raj Shrestha, Rudra P. K. Poudel & Binod Bhattarai

TEACHING EXPERIENCES

- **Teaching Assistant, Sixth Annual Nepal AI School (ANAIS) 2025** December 2025
Delivered lab sessions on geometric deep learning
- **Instructor, Competitive Programming Workshop, IEEE Pulchowk** July 2023
Prepared and delivered a 5-day workshop on competitive programming
- **Instructor, Software Fellowship, LOCUS 2023** June 2023
Prepared and delivered lectures on Data Structure and Algorithms
- **Instructor, Competitive Programming Workshop, Thapathali Campus** July 2022
Prepared and delivered a 6-day workshop on competitive programming

AWARDS AND ACHIEVEMENTS

- **International Math Olympiad**
Secured **1st position in National Math Olympiad** and represented Nepal at IMO 2019 (Bath, UK)
- **Other Olympiads**
 - **Winner of Various National Math and Physics Competitions:** Physics–Math Olympiad 2019 by Mechatrix, Maths Hunt 2018 by St. Xavier’s College
 - **Top 6** in National Physics Olympiad, 2019 pre-selection test
- **Competitive Programming**
 - Achieved **Global Ranking of 76 (1st Nationally)** Among 7091 Teams in IEEEExtreme Programming Competition 17.0
 - **Winner of Various National Coding Competitions:** LOCUS Codejam 2022, Yomari Coding Competition 2023, LOCUS Codejam 2023
 - Achieved Maximum Rating of **1793** on Codeforces
- **SmartBots** First Runner up at SmartBots Coding Challenge
Involved development of an efficient game playing bot for a card game, competition involved 94 teams nationwide
- **Scholarship** Merit-based full tuition waiver in undergrad based on entrance exam ranking

VOLUNTEER EXPERIENCES

- **Reviewer**, *Workshop on Data Engineering in Medical Imaging, MICCAI 2025* 2025
- **Mathematics Coordinator**, *Maths and Physics Society, Pulchowk Campus* 2022 – 2023

SELECTED ACADEMIC PROJECTS

- **Automata Toolkit:** Implemented functionalities to convert between DFAs, NFAs and regular expressions, minimize DFAs, and visually simulate DFAs and NFAs.
- **Interpreter for Custom Programming Language:** Implemented a general CFG parser using Earley Parsing; Full fledged implementation of functionalities typical of a high level programming language.
- **VHDL Simulator:** Implemented a subset of VHDL to simulate digital circuits programmatically; Added automatic circuit diagram generation from VHDL specifications.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, JavaScript
- **Frameworks:** PyTorch, NumPy, Matplotlib, Scikit-learn, MMDetection
- **Machine Learning:** Computer Vision, Generative Modeling, Reinforcement Learning
- **Other Tools:** Git, LaTeX, Shell Scripting, Docker, Slurm, Linux Environment